power

File: power.kicad_sch

mcu

File: mcu.kicad_sch

sensing

File: sensing.kicad_sch

vio

File: vio.kicad_sch

translators

File: translators.kicad_sch

connectors

File: connectors.kicad_sch

OOTY rev A — hierarchical schematic generated by scripts/build_schematic.py

Specification: SPEC.md — design notes and calculations: NOTES.md

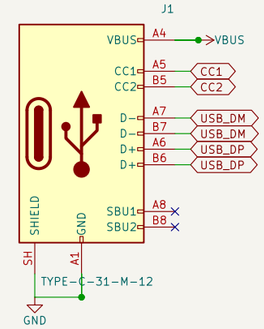
OOTY

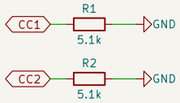
Sheet: /
File: OOTY.kicad_sch

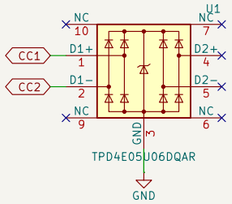
**Title: OOTY — USB Hardware-in-the-Loop board**

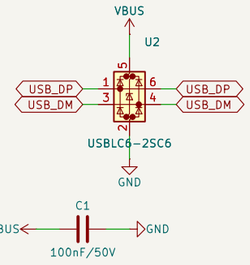
| Size: A4 | Date: 2026-10-20 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 1/7 |

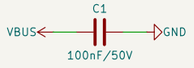
## USB-C input



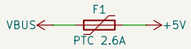



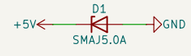



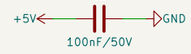



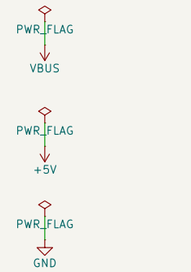




## VBUS protection

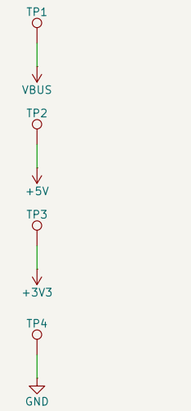


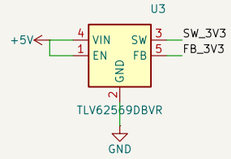




## Rail test points

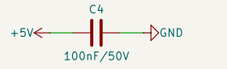



## 3V3 buck (TLV62569)



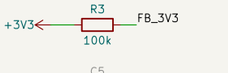






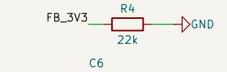






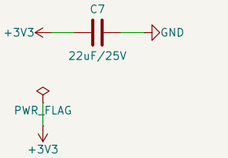




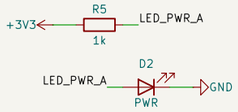




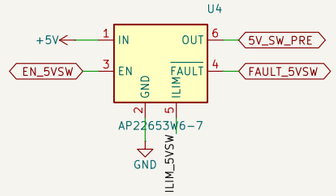


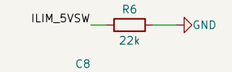


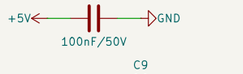

## Power OK LED




## Switch 5V_SW (switched, 1 A)



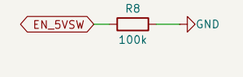




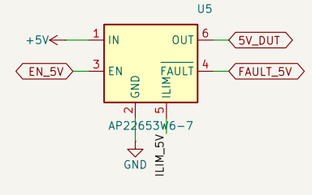


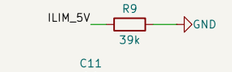







## Switch 5V fixed (0.5 A)

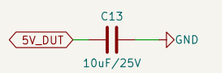


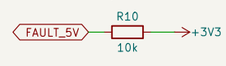









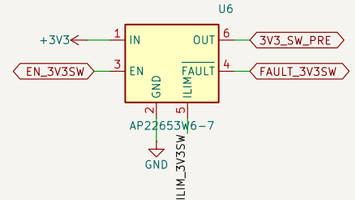



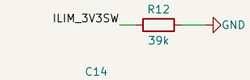







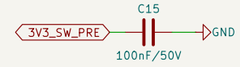
## Switch 3V3_SW (switched, 0.5 A)

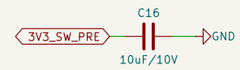


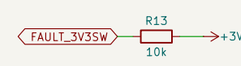



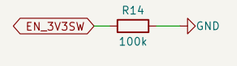




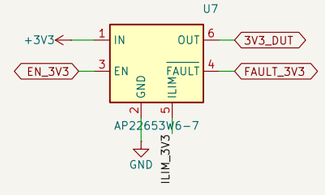


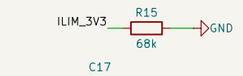



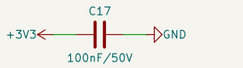



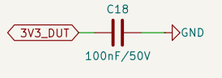



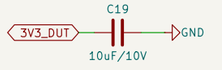

## Switch 3V3 fixed (0.3 A)

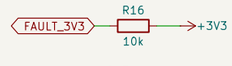






AP22653 ILIM (DS41186 p.10, VIN = 5 V): 22k -> 1.16 A typ, 39k -> 0.64 A typ, 68k -> 0.35 A typ. See NOTES.md.

Buck: Vout = 0.6 V x (1 + 100k/22k) = 3.327 V.

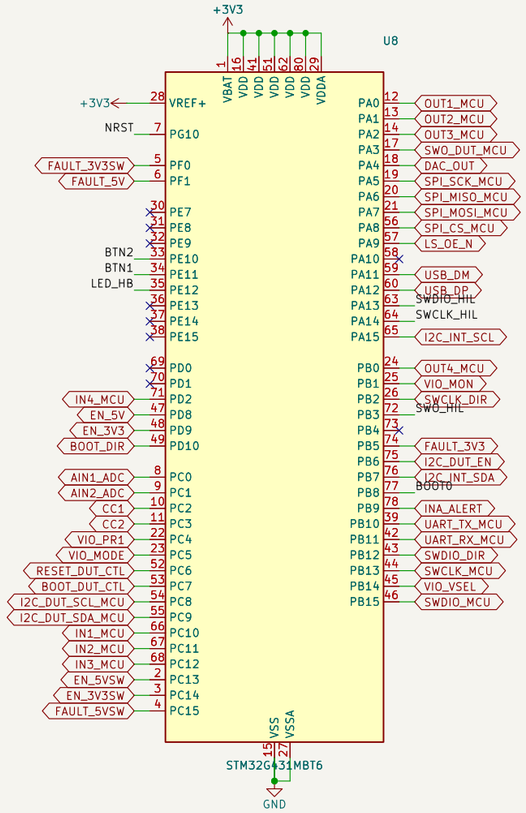
OOTY

Sheet: /power/
File: power.kicad_sch

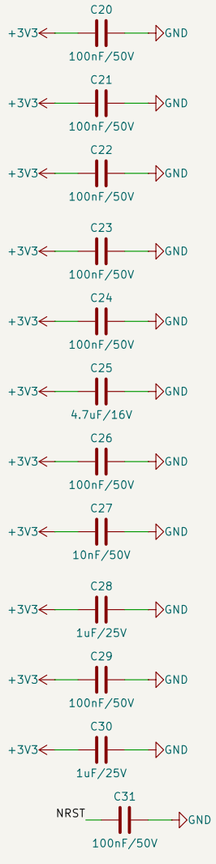
**Title: Power: USB-C, 3V3 buck, DUT rails**

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-20 | **Rev: A** |
| KiCad E.D.A. 10.0.7 | | Id: 2/7 |

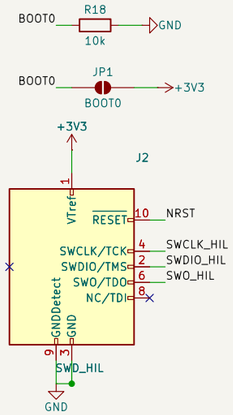
## STM32G431MBT6



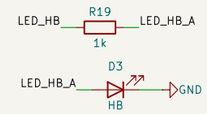

## Decoupling (DS12589 5.1.6)



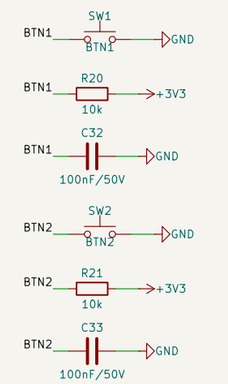

## Boot / HIL debug (SWD)




## Heartbeat LED




## User buttons



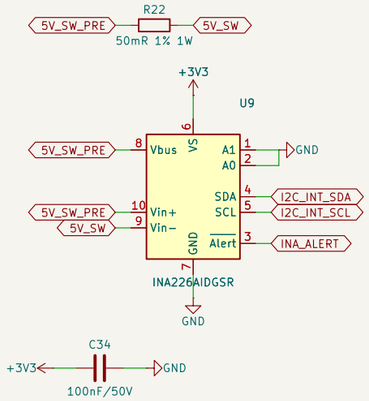

USB FS runs crystal-less (HSI48 + CRS). BOOT0 solder jumper: bridge it to force the system bootloader.

PB6 may carry a UCPD dead-battery pull-down at reset (PA9 = DBCC1 is pulled high); I2C_DUT_EN has an external pull-down anyway. Firmware sets PWR_CR3.UCPD1_DBDIS first.

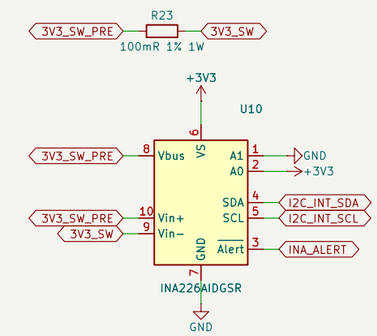
Pin choice follows the PCB floor plan (MCU rotated 180 deg): see NOTES.md.

| | | |
|---|---|---|
| OOTY | | |
| Sheet: /mcu/ | | |
| File: mcu.kicad_sch | | |
| **Title: MCU STM32G431MBT6** | | |
| Size: A3 | Date: 2026-10-20 | **Rev: A** |
| KiCad E.D.A. 10.0.7 | | Id: 3/7 |

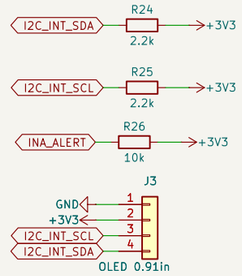
## Current monitor 5V_SW



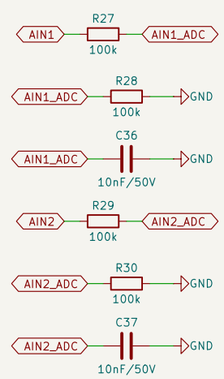

## Current monitor 3V3_SW



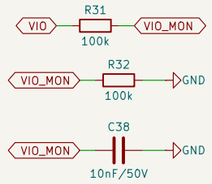

## Internal I2C + OLED



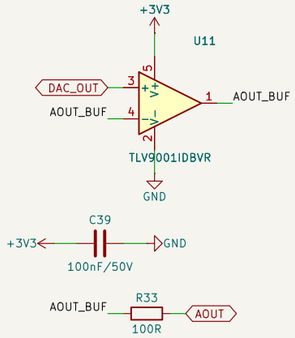

## Analog inputs (0–5.5 V)




## VIO monitor




## DAC output buffer

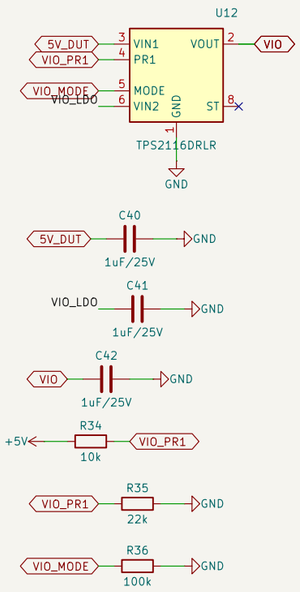



INA226 addresses: 5V_SW 0x40 (A1=A0=GND), 3V3_SW 0x41 (A0=VS). OLED 0x3C. INA228 is a drop-in replacement (same pinout).

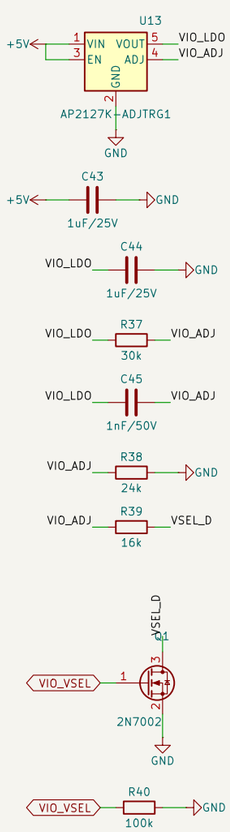
Shunts: Kelvin connection to IN+/IN− required at layout.

| | | |
|---|---|---|
| OOTY | | |
| Sheet: /sensing/ | | |
| File: sensing.kicad_sch | | |
| **Title: Current sensing, OLED, analog I/O** | | |
| Size: A3 | Date: 2026-10-20 | **Rev: A** |
| KiCad E.D.A. 10.0.7 | | Id: 4/7 |

## VIO selection (TPS2116 mux)




## VIO LDO 1.8 / 3.3 V (AP2127K-ADJ)

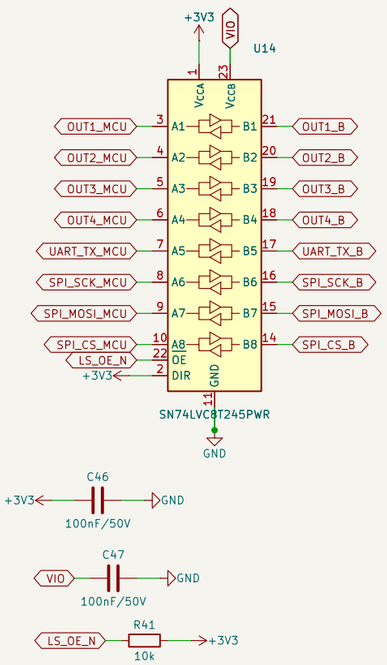



TPS2116: MODE=1 PR1=1 -> 5V_DUT ; MODE=1 PR1=0 -> LDO ; MODE=0 PR1=1 -> off (external VIO from J_PWR/J_DBG). MODE=0 PR1=0 is forbidden (selects 5 V).

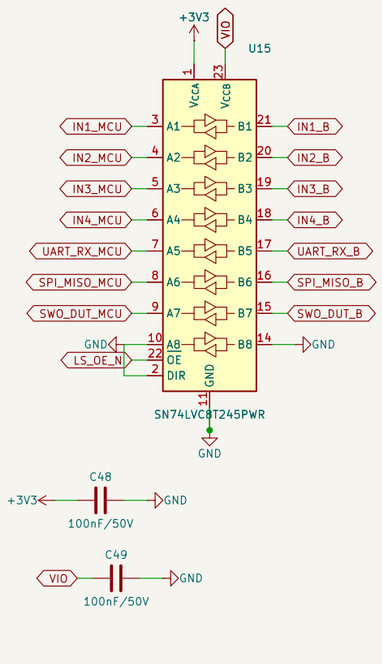
LDO: Vout = 0.8 x (1 + 30k/24k) = 1.80 V ; VIO_VSEL high adds 16k in parallel -> 3.30 V.

| | | |
|---|---|---|
| OOTY | | |
| Sheet: /vio/<br>File: vio.kicad_sch | | |
| **Title: VIO: 1.8 / 3.3 / 5 V or external** | | |
| Size: A3 | Date: 2026-10-20 | **Rev: A** |
| KiCad E.D.A. 10.0.7 | | Id: 5/7 |

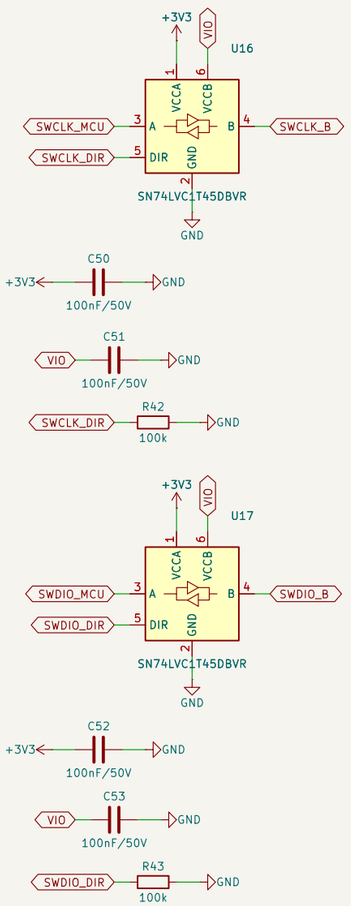
## Outputs 3V3 -> VIO (DIR = A to B)



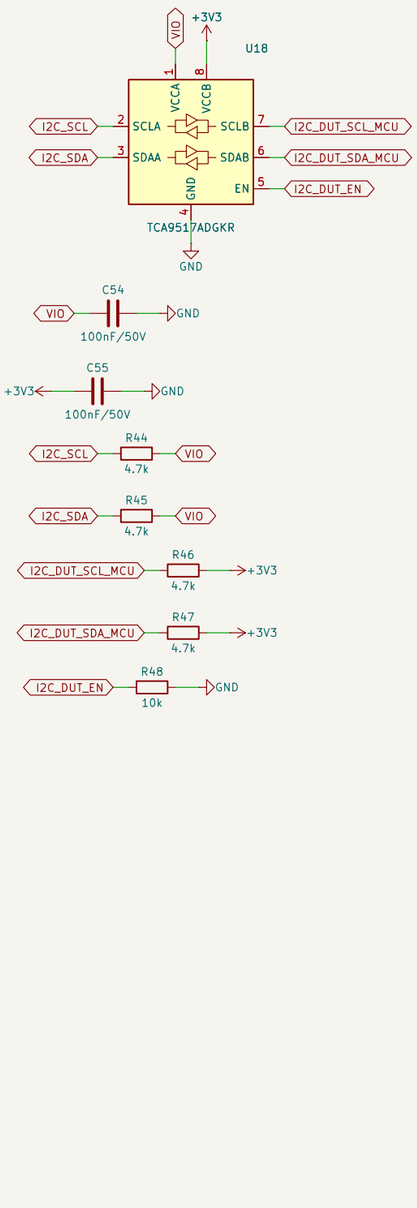

## Inputs VIO -> 3V3 (DIR = B to A)



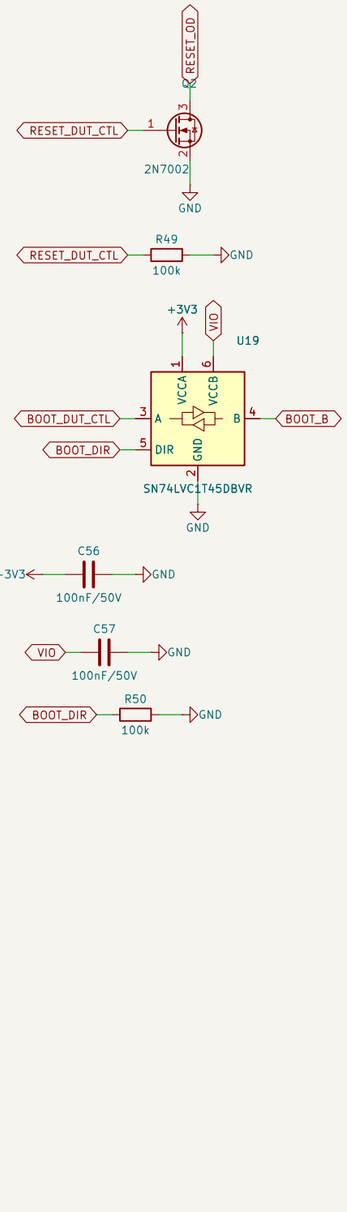

## SWD probe



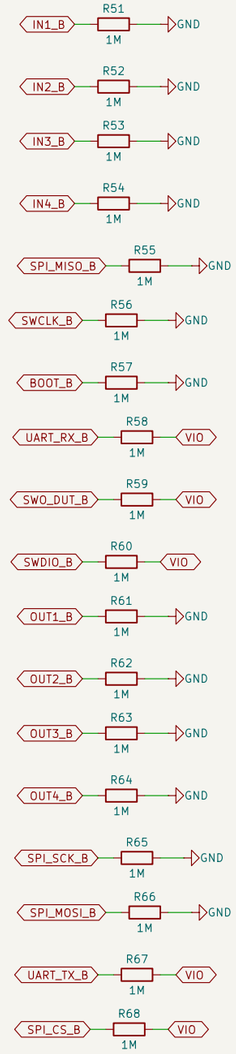

## DUT I2C (A = DUT)




## RESET (OD) / BOOT (3-state)



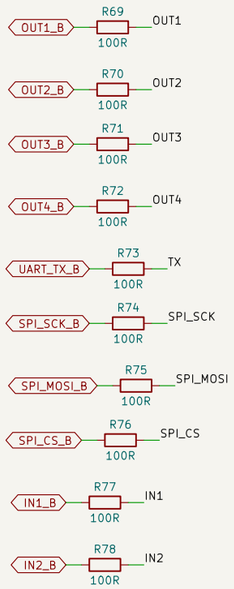

## Input bias 1M



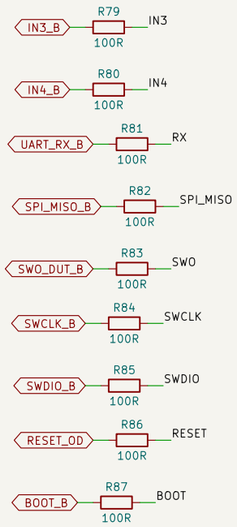

All translators are Hi-Z until firmware drives LS_OE_N low / DIR high / I2C_DUT_EN high.

| | | |
|---|---|---|
| OOTY | | |
| Sheet: /translators/ | | |
| File: translators.kicad_sch | | |
| **Title: Level translators 3V3 <-> VIO** | | |
| Size: A3 | Date: 2026-10-20 | **Rev: A** |
| KiCad E.D.A. 10.0.7 | | Id: 6/7 |

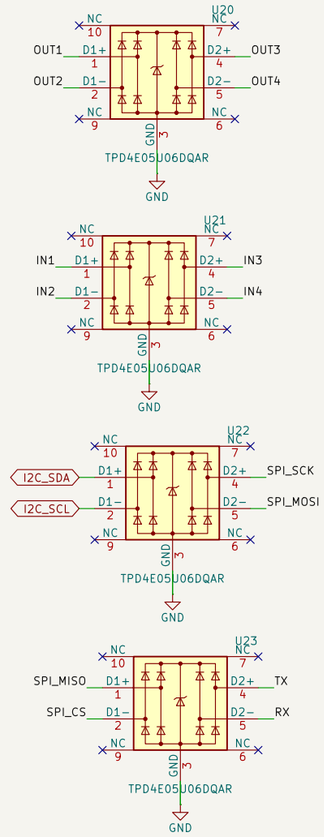
## Series resistors



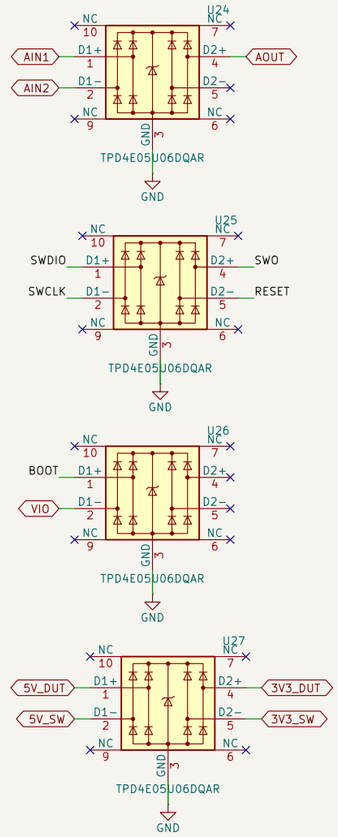

## Series resistors (cont.)



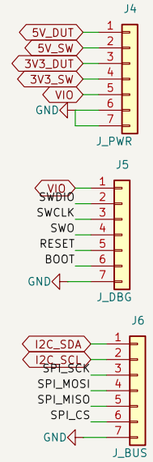

## ESD (TPD4E05U06)



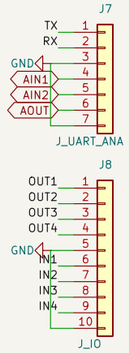

## ESD (cont.)




## DUT headers (2.54 mm)




## DUT headers (cont.)




## Mounting holes



Signal path: header -> TVS -> 100R series -> translator / circuit.

DUT rails and VIO are probed on J_PWR / J_DBG pins (no separate test points).

| | | |
|---|---|---|
| OOTY | | |
| Sheet: /connectors/ | | |
| File: connectors.kicad_sch | | |
| **Title: DUT connectors and ESD** | | |
| Size: A3 | Date: 2026-10-20 | **Rev: A** |
| KiCad E.D.A. 10.0.7 | | Id: 7/7 |